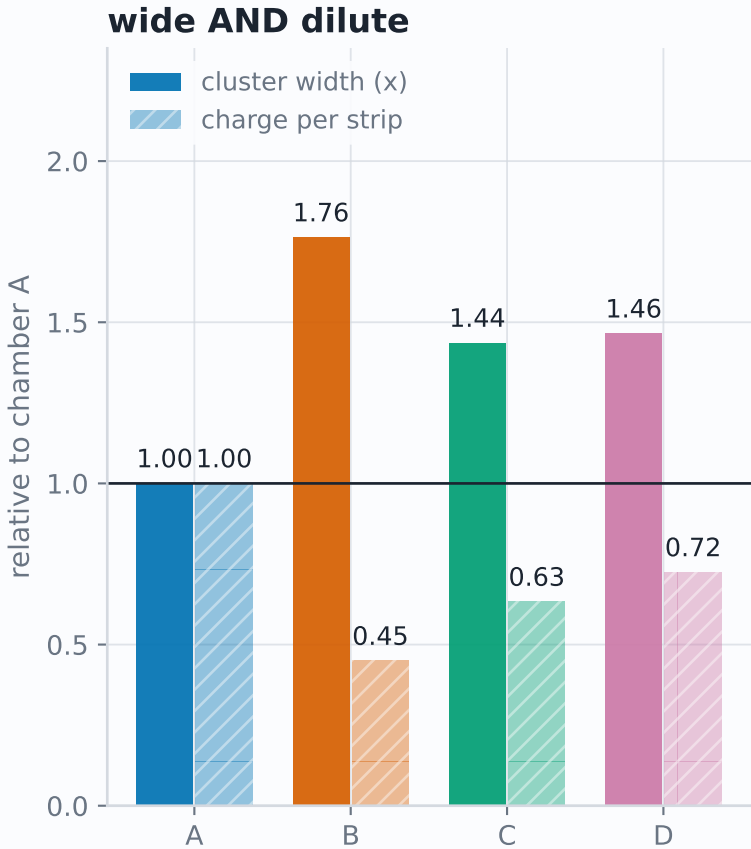
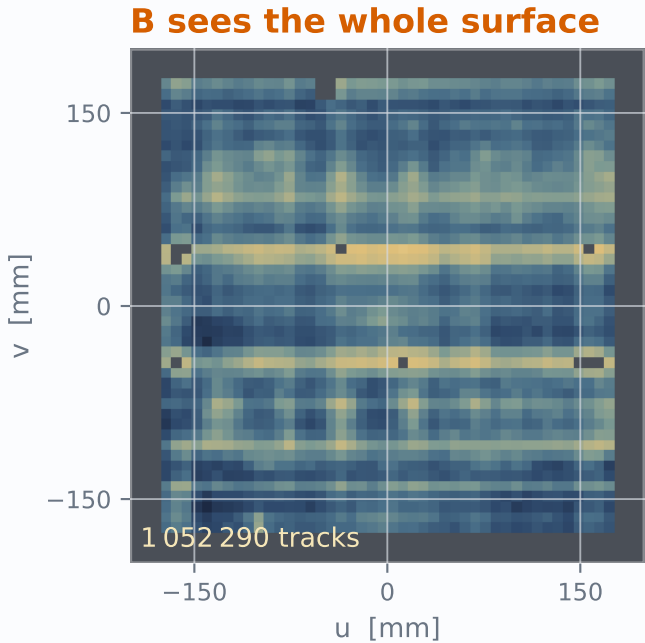


# Chamber B is a field-cage fault, not a dead detector

PRELIMINARY

it has no field-shaping ring chain, so it carries no angle — and everything else about it works



## the cause, in the HV monitor

| arm | drift I  | resistive I | ring chain |
|-----|----------|-------------|------------|
| A   | 0.180 μA | 0.088 μA    | yes        |
| B   | 0.000 μA | 2.136 μA    | NONE       |
| C   | 0.180 μA | 0.013 μA    | yes        |
| D   | 0.180 μA | 0.800 μA    | yes        |

700 V / 0.180 μA = 3.89 GΩ = 3 × 1.30 GΩ — the degrader divider, and nothing else. B draws zero because B has no divider to draw it. Without the rings the drift field fringes, so there is no clean time↔depth ladder and no angle — with the cathode powered exactly as it should be.

The amplification stage is untouched, so B still measures position and time. That is what the map on the left is, and it is where B belongs.

Cluster shape: median over all gated clusters, the basis `chamber\_b.py` established on run\_145 and this reproduces across the campaign. A fringing field spreads the same charge over more strips, so the prediction is a wide, DILUTE cluster and not a weak one — and that is exactly what separates B: widest of the four, and clear of the other three on density. The map is the bystander sample, which is the only one B can enter. B’s resistive channel drawing 2.136 μA against A’s 0.088 is a separate anomaly and is still unexplained.